

JAMES OGBORNE



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github.com/James-the-coder

A Computer Science and AI graduate from the University Of Bath with 15 months of industry experience as an AI Data & Software Engineering intern at an award-winning medical device startup. Specialising in Python-based data pipeline engineering, MLOps, and deep reinforcement learning, with hands-on research experience implementing state-of-the-art algorithms in robotic simulation environments. Eager to apply and expand these skills in a high-impact AI/ML engineering or research role

EXPERIENCE

AI Data & Software Engineering Intern

July 2024 - September 2025

Afon Technology Ltd. - Medical Device Startup (Award-Winning)

13-month industrial placement. Operated across data infrastructure, internal analytics, and machine learning research in a multi-disciplinary team.

- Developed a modular, end-to-end machine learning pipeline for glucose prediction from wearable sensor data, with configuration-driven model selection, dataset processing, and training, enabling rapid, reproducible experimentation across the research team.
- Investigated deep learning approaches for predicting glucose levels from wearable sensor data collected during clinical trials.
- Iteratively designed and maintained a suite of internal Python tools to support data analysis.
- Conducted exploratory data analysis to identify high-quality data regions for model training, and independently diagnosed hardware and firmware issues from device data, directly contributing to the discovery and resolution of firmware bugs.
- Regularly presented software functionality, data analysis findings, and research progress to both technical and non-technical stakeholders, including end users of the tools.

AI Data & Software Engineering Intern

July - September 2023

Initial 2-month summer internship with the same employer; responsibilities consistent with the extended role above. Performance during this period led to the offer of the 13-month industrial placement.

EDUCATION

BSc (Hons) Computer Science with Artificial Intelligence

2022 - 2026

University Of Bath - With Industrial Placement Year - 2:1 (69%)

Final year dissertation: A Comparative Analysis of Intrinsically Motivated Deep Reinforcement Learning Methods in Robotic Arm Manipulation Tasks.

- Implemented SAC+HER from scratch in Python and augmented it with two intrinsic motivation modules (ICM and RND) to comparatively evaluate their effect on exploration and sample efficiency in sparse reward robotic pick-and-place tasks within MuJoCo on a multi-node university GPU cluster.
- Relevant modules include: Computer Vision, Natural Language Processing, Reinforcement Learning, and Entrepreneurship.

Monmouth School for Boys

2015 - 2022

A levels: Further Mathematics (B), Mathematics (A), Physics (A), Computer Science (A)

GCSEs: Seven grade 9s, three grade 8s

TECHNICAL SKILLS

Languages: Python (advanced), SQL, Java, C, JavaScript, Bash

ML / AI: PyTorch, TensorFlow, scikit-learn, pandas

Simulation: MuJoCo (robotics manipulation environments)

Infrastructure: Docker, Linux/Ubuntu, Git, GPU cluster operation

Data Engineering: Pipeline design, automated ETL workflows

Other: React Native, Node.js, Google Firebase, MySQL, Raspberry Pi, Arduino

ACTIVITIES AND ACHIEVEMENTS

- Pursuing a self-directed interest in quantum computing, having completed Stanford Online's Quantum Mechanics for Scientists and Engineers 1 course, and two IBM courses in Basics of Quantum Information and Fundamentals of Quantum Algorithms.
 - Engineering Education Scheme Wales (EESW) Award: collaborative engineering project solving a real-world industry problem.
 - Sixth Form Computer Science Award & Computer Science Prefect, Monmouth School for Boys.
 - Bronze Physics Olympiad, Bebras Distinction, Bronze Duke of Edinburgh Award.
 - Survive & Save Bronze Award, Royal Lifesaving Society UK (2020)
 - Keen rock climber.
 - Volunteered as a programming mentor at a school computer science club (teaching Raspberry Pi) and youth tennis coach.
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